

Research on an Automatic Sorting System Based on Machine Vision

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Abstract—The sorting and handling of materials is an essential part of the automated production line. However, the traditional sorting method cannot meet the current development needs due to low efficiency. This paper proposes an automatic sorting system by pneumatic robots with machine vision technology. The system is composed of machine vision unit, automatic conveying unit, pneumatic robot unit and controller unit. The machine vision software Halcon is selected to recognize the image accurately, and a sorting execution device is designed which is combined with pneumatic manipulator and pushing mechanism. Programmable controller (PLC) is selected as the controller, and the hardware and software of the control system are designed. Finally, the system is simulated and the results show that the system can quickly realize the sorting of materials of different shapes and colours, and improve the efficiency and accuracy of sorting.

Keywords—Machine vision, sorting, pneumatic robot, PLC

I INTRODUCTION

The sorting and handling of materials is an essential part of the automated production line. In the past, the sorting and handling of materials was manual sorting, which required a large number of workers to work at the same time and didn't ensure the health of the workers and their work and rest time. Moreover, the manual sorting efficiency was low, and it was subject to many interference factors, which may be incorrectly sorted or damaged materials. With the development of automation technology, manipulators gradually replaced manpower and began to complete repetitive tasks such as material palletizing, assembly, sorting and handling on automated production lines, which greatly improved the efficiency of sorting and handling, extended working hours, and reduced enterprise costs. It is helpful to improve the competitiveness of enterprises. China Post has launched a cross-belt sorting system to realize rapid sorting, sorting and delivery of mail [1]. Zhaorui Li proposed an automatic cigarette sorting system to optimize replenishment [2]. Roodbergen and Koster optimized the path optimization strategy of multiple channels, multiple commodities and multiple sorting and handling paths in multiple sorting processes [3]. Elizabeth Jewkes proposed a sorting and handling system which is composed of five units [4]. Through scientific calculation and analysis, different orders and different conveyor belts form a sorting and handling mechanism to improve the efficiency of sorting goods. LoonChing Tang selected and designed a

conveyor of the material sorting system, and through the analysis of the material movement time, speed and direction, the basic method of automatic sorting and transportation was proposed [5]. As the market has higher and higher requirements for automation, it is necessary to use automatic equipment to complete more complex sorting and handling actions. Industrial robots can achieve motion trajectory planning and manipulator actions through programming, which improves the sorting efficiency. However, the devices can perform fixed actions only, which have strict requirements on the location of material. With the market's demand for high efficiency and intelligence, industrial robots equipped with machine vision systems are more intelligent and can handle changing situations and complex environments.

II THE COMPOSITION OF THE SYSTEM

The system configuration scheme is shown in Figure 1, which mainly includes four components, namely machine vision unit, automatic conveying unit, pneumatic robot unit and total control unit. The controller uses a programmable controller (PLC), the machine vision unit uses a CCD camera to connect to the PC for graphics processing, the pneumatic robot uses a solenoid valve to control its action, and the conveyor unit uses a belt-type transport equipment which is driven by the three-phase asynchronous motor with frequency converter.

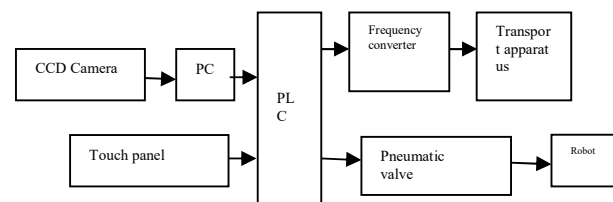


Figure 1. The composition of the system

The structure diagram of the system is shown in Figure 2.

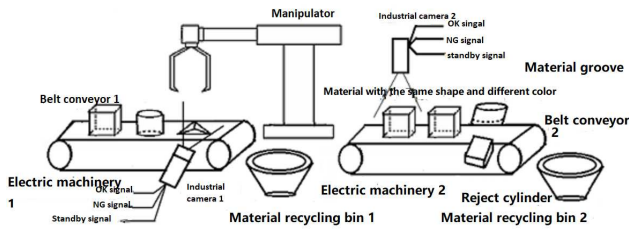


Figure 2. The structure diagram of the system

Industrial cameras process, analyze, and recognize material information through algorithms such as image preprocessing, mathematical morphology, edge detection, threshold segmentation, and Blob analysis, and pass the scan and output of the camera to the image capture card, and then input to the vision software in the computer to obtain measurement results or logical control values. The vision system will send multiple signals to the PLC, and the most important ones are the OK signal, NG signal and standby signal. When the qualified material is visually identified, analyzed, and detected, the vision system will send an OK signal to the PLC, and the unqualified material vision system will send an NG signal to the PLC, and the PLC controls the mechanical manual and rejects mechanism actions through the output signal. The standby signal can control the start and end of the visual recognition of the visual system. In this system, the industrial camera 1 recognizes and determines the OK signal and NG signal based on shape recognition. For example, a circle is an OK signal, and the others are NG signals. The industrial camera 2 recognizes and determines the OK signal and NG signal based on Color recognition, such as yellow is an OK signal, other colors are NG signals. A suitable manipulator was installed between the two conveying and sorting mechanisms. The PLC control is divided into automatic control and manual control for grabbing and conveying materials. The PLC control program is used to realize the extension, retraction, ascending and descending of the manipulator, Turn left, turn right, and grip and loosen the claw, so as to accurately complete the grasping and conveying of materials. When the industrial camera 1 inputs the OK signal, it starts to drive the manipulator to move.

When the material is placed on the belt, the PLC starts the motor, and the motor runs at a certain frequency to drive the belt to rotate. When the industrial camera 1 inputs the OK signal or the industrial camera 2 inputs the NG signal, the inverter 1 or 2 will make the motor 1 or 2 run at low speed to prepare for the grasping, handling and sorting of materials. The end of the belt conveyor is equipped with a material recovery bucket to collect the sorted materials in a unified manner.

III DETAILED DESIGN OF THE SYSTEM

A. Design of pneumatic actuator

This system installs a pneumatic manipulator between the conveying and feeding mechanism and the conveying and sorting mechanism. The spatial position of the material

is changed through the grasping and conveying of the manipulator. The pneumatic manipulator is selected because of its simple structure, lighter weight, and rapid action. The advantages are that the gas source is convenient, the cost is low, and the maintenance is convenient. This design requires the manipulator to complete the main actions such as ascending and descending, extending and retracting, turning left and right, and grasping and releasing the manipulator claw.

In order to meet the action requirements of the manipulator, the pneumatic circuit system of the pneumatic manipulator was designed. The schematic diagram of the pneumatic system of the manipulator is shown in Figure 3.

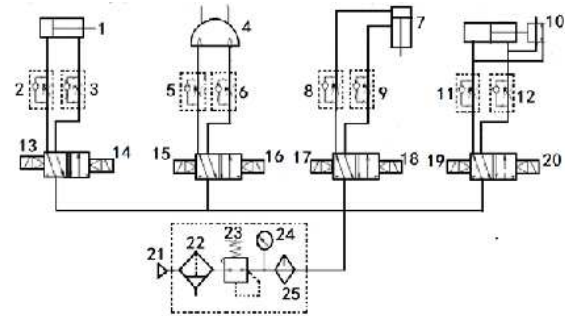


Figure 3. Schematic diagram of pneumatic system

In the pneumatic manipulator control system, when materials appear in the conveying and feeding mechanism, the industrial camera will send out an OK signal for qualified materials and NG signals for unqualified materials through visual recognition. After receiving the OK signal, the pneumatic manipulator starts to work. The pneumatic manipulator starts to extend the arm. After the front extension limit magnetic sensor detects the signal, the manipulator arm starts to descend. After the lower limit magnetic sensor detects the signal, the pneumatic finger of the pneumatic manipulator starts to grasp. Take the material, when the pneumatic finger clamping limit sensor detects the clamping signal, the pneumatic manipulator arm starts to rise. After the upper limit magnetic sensor detects the signal, the manipulator begins to retract. After the retract limit magnetic sensor detects the signal, the pneumatic manipulator starts to rotate to the right. After the rotating cylinder clockwise limit magnetic sensor detects the signal, the arm begins to extend. After the front extension limit magnetic sensor detects the signal, the pneumatic manipulator arm begins to descend. After the lower limit magnetic sensor detects the signal, the pneumatic finger of the pneumatic manipulator starts to release the material. When the pneumatic finger release limit sensor detects the release signal, the pneumatic manipulator arm starts to rise. After the upper limit magnetic sensor detects the signal, the pneumatic manipulator starts to rotate to the left and the rotating cylinder reverses. After the hour hand limit magnetic sensor detects the signal, the manipulator starts to wait for the next material to be in place and then the manipulator performs a cycle action again. The action of the

entire manipulator can be controlled automatically and manually. When the automatic control is performed, the pneumatic manipulator will work repeatedly. In manual control, after manually clicking the start button, the pneumatic manipulator will complete an action, and then stop the movement after completion.

B. Belt conveyors

The motor is used to drive the belt to convey materials, which can control the conveying direction and conveying speed of the conveying system. The belt conveyor 1 in the system is used to convey, sort and grab materials. It is composed of a motor, a frequency converter, a manipulator, a vision system and a material recovery barrel. It can transport materials and drive the manipulator to grasp by visual identification and detection. To take the material, the position of the industrial camera of the vision system is fixed next to the conveyor line. There is a fixed field of view that will not change with the movement of the manipulator. This improves the position accuracy during the visual recognition process and will not change the field of view range with the movement of the manipulator. The input signals of the vision system are OK signal, NG signal and standby signal. The material with the same shape, such as a circle, is judged as an OK signal, and the other shapes are judged as an NG signal. The manipulator grabs the material with the OK signal, and the material with the NG signal falls into the material recovery bucket, thereby sorting the shape of the material.

C. Machine Vision Unit

Hardware design of machine vision. The light source directly affects the image quality of the machine vision system during image acquisition. After comparing the characteristics of various light sources, it is found that the fluorescent lamp has large brightness, good light source uniformity, and long service life, which can make the measured image and the surrounding environment have a greater contrast. After adjusting the position of the light source and the target object, a good quality image can be formed, so it was decided to choose fluorescent lamp as the light source of the system.

The camera can determine the detected image resolution and image grey level, and also affect the motion of the system. Therefore, choosing the right camera is particularly important. Its main characteristics include resolution, maximum frame rate, exposure mode, dynamic range, etc. After comprehensive consideration, the UC-930CI camera of UNIQ was selected. The sensor type of this product is CCD, and the horizontal and vertical resolutions are both large.

The image capture card converts the image signal into a signal that the computer can recognize and receives, and inputs it to the storage unit in the computer. After computer processing, it can judge the basic information of the material and the position information of the object movement. The model of MV-GigE-2 high-speed digital image capture card is chosen. The characteristics of the image capture card are that it conforms to the PCI-E bus

standard, has high transmission efficiency, supports multi-core operating systems, facilitates connection with computers, and supports multiple image data transmission formats.

Machine vision software. Halcon has a rich vision development environment and is equipped with many basic vision algorithm packages. The development time is short to obtain image stabilization. Halcon relies on a variety of algorithms and is widely used in industrial production, medical imaging, and electronic manufacturing industries.

Because there is no accurate method to define the shape of the image target, it will choose to use mathematical methods to define shape parameters to describe the shape. First, the RGB image is greyed out to reduce the interference of colour on shape recognition. Then select a specific area, and build a model through image binarization, threshold segmentation, pattern matching and Blob analysis. Finally, the detected model is displayed and expressed. Use `gen_rectangle()` to generate a rectangle, `reduce_domain()` to reduce the image domain. After replacing the image with a grayscale image, use `threshold()` to select the threshold of a specific range, use `area_center` to find the centre point coordinates of the RegionFillUp area, and use the image to extract the region when the image is processed, there are often several similar regions, and the specified region needs to be selected according to some special characteristics of the region. The `gen_rectangle1` operator is used to generate a rectangle. The five parameters are the generated rectangular area. The row of the upper left corner indicates the row, the column of the upper left corner indicates, the row of the lower right corner indicates and the column of the lower right corner indicates. When the straightness of the RegionTrans area is greater than 0 and less than 0.65, the system judges it as a triangular area. When the straightness is greater than 0.80, the system judges it as a rectangular area. When the roundness is greater than 0.90, the system judges it as a circular area. The algorithm is shown in Figure 4.

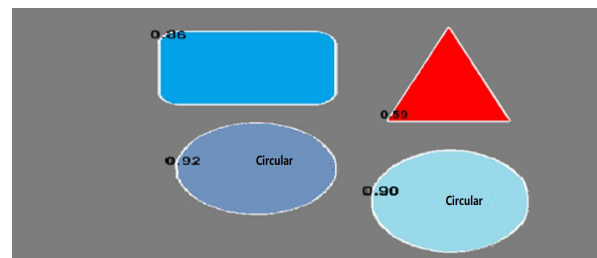


Figure 4. Shape recognition

RGB color space is based on three colors of red, green, and blue, and multiple colors are directly superimposed on each other. The HSV color space performs a variety of parameter digitization according to the performance characteristics of the color, where H (hue) represents the position of the color in the spectrum, S (saturation) represents the closeness of the color in the spectrum, and V (brightness) indicates the intensity of the color intensity.

First collect the image and split the image into three channels, namely R, G, B, and convert RGB to HSV through `trans_rom_g`. The color of the object is represented by chroma (H), saturation (S) and brightness (V). At this time, there are six images of R, G, B, H, S, V in the variable window. As shown in Figure 5, the right image is the H variable grayscale image of the HSV image. The corresponding area is selected after the brightness (S) binarization through the threshold function. Then select the picture according to the gray value of the chroma (H) of the area. Cut, break into different connected domains, perform binarization, extract, select the corresponding area, perform the closing operation to extract the corresponding color, and finally cut out the corresponding area image, and modify the threshold range and shape Cut, select the corresponding color, and finally perform color recognition, as shown in Figure 6.

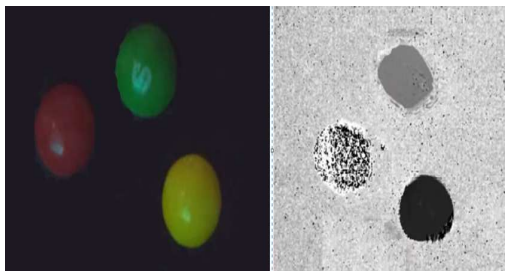


Figure 5. H-variable grayscale image

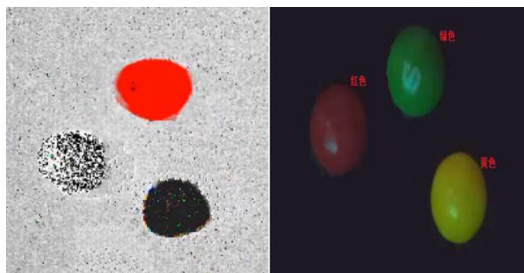


Figure 6. Brightness and chroma range

IV Control system design and simulation test

The control unit of the system is composed of PC and PLC. PC is the processor of machine vision unit, and PLC is the overall controller of the system. The PLC is connected to external equipment through input and output modules, including pneumatic manipulators and solenoid valves that eliminate the pneumatic circuit of the cylinder, frequency converters, and machine vision processor. After the PLC is started, the system will perform power-on initialization, and then perform self-diagnosis, check the conditions between the various units in the PLC in turn, and then communicate with external devices, input and output processing signals through the I/O port, and proceed with the program in sequence control.

The ladder diagram of this system can be divided into three aspects, namely the automatic working mode of the pneumatic manipulator, the manual working mode of the pneumatic manipulator and the sorting mode of the belt conveyor. The action of the entire manipulator can be controlled automatically and manually. When the automatic control is performed, the pneumatic manipulator will work repeatedly. The sorting process of the belt conveyor is divided into two parts: belt conveyor 1 and belt conveyor 2. In the belt conveyor 1, the industrial camera 1 recognizes the shape through image processing, and the recognized OK signal is transmitted to the PLC, and the PLC drives the manipulator to move. Change the frequency converter to a low-speed gear to facilitate the grabbing of materials. The materials with NG signal fall into the material recovery barrel 1. In the belt conveyor 2, the industrial camera 2 performs image processing and color recognition, judges the unqualified color material as NG signal, and inputs it to the PLC. The PLC drives the cylinder to remove the unqualified material and pushes the unqualified material into the material tank. If the judgment is OK, the signal material falls into the material recovery bin 2. After sorting twice, you can get materials with specific colors and specific shapes.

Configuration software is usually also called configuration monitoring system software in the industry. The software can perform logical and digital processing on industrial sites by connecting to PLC. The information displayed by the system is generally embodied by graphics and animation, based on the performance of images and animations. Then the soft of Kingview can be used to test the monitoring and control tasks of the industrial site. When using the Kingview software for analysis, make the hardware control plan and the input and output points of the hardware equipment, and analyze according to the working requirements of the system and the connected external equipment. The test results show that the system can accurately complete the sorting and handling of materials, with high efficiency and good stability. The system can be popularized and used in various measurement aspects such as material defect detection and quality inspection, which improves production efficiency, increases work accuracy, and improves the core competitiveness of enterprises.

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